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Summary

Software engineer and data scientist with experience in QA automation, data engineering, backend development, and applied machine learning. Currently pursuing an M.Sc. in Data Science at Universität Potsdam and building Python, CI/CD, SQL monitoring, and LLM-powered developer tooling at MBition.

Education

Universität Potsdam <i>Master of Science in Data Science</i> Current GPA: 1.7; coursework focused on machine learning, deep learning, probability, and Bayesian statistics.	October 2024 – Present <i>Potsdam, Germany</i>
The German International University - Berlin <i>Semester Abroad Program</i>	September 2021 – March 2022 <i>Berlin, Germany</i>
The German University in Cairo <i>Bachelor of Science in Computer Science and Engineering</i> - Graduated with Highest Honors (Grade: A+). - Bachelor thesis: Brain Hemorrhage Detection from CT Images using Deep Learning (Grade: A+).	September 2017 – July 2022 <i>New Cairo, Egypt</i>

Skills

Languages: Python, Java, SQL, JavaScript, HTML/CSS
Data & ML: Pandas, Spark, Dask, TensorFlow, OpenCV, Grafana, Hadoop
Backend & Cloud: Flask, REST APIs, Docker, AWS, PostgreSQL, MySQL, MongoDB, SQL Server, Hibernate, Maven, JUnit, Linux, Git
Spoken Languages: Arabic (native), English (fluent), German (B1)

Experience

MBition, Mercedes-Benz Group <i>Working Student - Test and Integration Strategy Team</i> - Built Python-based QA automation in CI/CD pipelines to validate repository compliance and reduce manual review effort. - Designed an LLM-powered internal assistant and SQL/Grafana monitoring workflows to improve developer QA processes.	Dec 2024 – Present <i>Berlin, Germany</i>
B_labs – Btech <i>Data Engineering Intern</i> - Engineered ETL pipelines for high-volume data using AWS services and cloud-based ingestion workflows. - Contributed to migration planning from a data warehouse to an AWS S3-based data lakehouse.	Jul 2024 – Sep 2024 <i>Cairo, Egypt</i>
Future of Egypt for Sustainable Development <i>Administrative and Technical Assistant - Military Service</i> Maintained the archive system and provided technical support for office systems, equipment, and network operations.	Feb 2023 – Feb 2024 <i>Cairo, Egypt</i>
German International University (GIU) <i>Computer Science Teaching Assistant</i> - Taught Data Structures and Algorithms using Java and led tutorial sessions for undergraduate students. - Created Python-based introductory computer science tutorials for first-year students.	Sep 2022 – Jan 2023 <i>Cairo, Egypt</i>
Aman for E-payment, Raya Holding <i>Java Backend Developer Intern</i> - Designed RESTful APIs and middleware blueprints using Apache Camel for system-to-system integration. - Worked with REST, JSON, XML, SOAP, Hibernate, and PostgreSQL in a Java backend environment.	Aug 2021 – Oct 2021 <i>6th of October, Giza, Egypt</i>

Projects

Brain Hemorrhage Detection with Deep Learning | GitHub **2021 – 2022**

- Built a YOLOv5 and PyTorch pipeline for brain hemorrhage detection from CT images with OpenCV preprocessing and a Flask interface for inference.
- Completed as a bachelor thesis project, demonstrating applied deep learning, computer vision, and model deployment skills.

Advanced Computer Lab – Staff Management System | GitHub **2022**

- Developed a full-stack HR management system using Node.js, Express, React, MongoDB, and JWT-based role-based access control for five user roles.
- Containerized the multi-service application with Docker to simplify setup, local testing, and integrated execution.

Customer Churn Prediction | GitHub **2021**

- Built an end-to-end machine learning workflow on 7,000+ telecom customer records with feature engineering, preprocessing, and model comparison.
- Evaluated Logistic Regression, Random Forest, and XGBoost models and identified churn drivers using feature-importance analysis.

Lightweight Database Engine | GitHub **2020**

- Implemented a Java database engine with B+ tree and R-tree indexing for efficient lookup, spatial search, and range queries.
- Built paged storage, clustering keys, and serialization mechanisms for persistent data management.